**1** **(a)** **(a)** 4 **[B1]**

**(b)** Orders the data: $4, 4, 5, 7, 8, 9, 12$ **[M1]**

7 **[A1]**

**(c)** $12 - 4 = 8$ **[B1]**

*[Total Marks: 3]*

**2** **(a)** 1 **[B1]**

**(b)** Ordered list: $0, 0, 1, 1, 1, 1, 2, 2, 3, 4$ **[M1]**

1 **[A1]**

**(c)** $\dfrac{2+0+1+4+1+0+3+1+2+1}{10} = \dfrac{15}{10}$ **[M1]**

1.5 **[A1]**

*[Total Marks: 5]*

**3** **(a)** 16 **[B1]**

**(b)** 21 **[B1]**

**(c)** Ordered list seen: $5, 8, 10, 13, 14, 16, 18, 20, 22, 24, 26$ **[M1]**

12 **[A1]**

**(d)** $(14 + 20 + 8 + 24 + 16 + 5 + 26 + 13 + 22 + 10 + 18) \div 11$ or $176 \div 11$ **[M1]**

16 **[A1]**

*[Total Marks: 6]*

**4** **(a)** 7 **[B1]**

**(b)** $12 - 7 = 5$ **[B1]**

**(c)** 8 **[B1]**

**(d)** $(7 + 7 + 7 + 8 + 10 + 12 + 12) \div 7$ or $\dfrac{63}{7}$ **[M1]**

9 **[A1]**

**(e)** $\dfrac{63 + y}{8} = 9.5$ or $9.5 \times 8$ **[M1]**

$76 - 63 = 13$ **[A1]**

*[Total Marks: 7]*

**5** **(a)** 15 **[B1]**

**(b)** **(i)** Ordered list: $8, 10, 11, 13, 14, 15, 15, 18, 21, 23, 50$ **[M1]**

15 **[A1]**

**(ii)** Lower quartile $Q_1 = 11$ and upper quartile $Q_3 = 21$ **[M1]**

$21 - 11 = 10$ **[A1]**

**(c)** $\dfrac{8+10+11+13+14+15+15+18+21+23+50}{11} = \dfrac{198}{11}$ **[M1]**

18 **[A1]**

**(d)** The value 50 is an extreme outlier that increases the mean, whereas the median is not affected by extreme values **[B1]**

*[Total Marks: 8]*

**6 (a)** $4 + x + 18 + y + 6 = 50$ simplifying to $x + y = 22$ **[B1]**

**(b) (i)** $\sum fm = 4(5) + 15x + 18(25) + 35y + 6(45) = 15x + 35y + 740$ **[M1]**

$\dfrac{15x + 35y + 740}{50} = 27$ giving $15x + 35y + 740 = 1350$ **[M1]**

$15x + 35y = 610$ dividing by 5 to give $3x + 7y = 122$ **[A1]**

**(ii)** Eliminating one variable, for example $(3x + 7y) - 3(x + y) = 122 - 66$ giving $4y = 56$ **[M1]**

$y = 14$ **[A1]**

$x = 8$ **[A1]**

**(c)** $20 < h \leq 30$ **[B1]**

**(d) (i)** $30 < h \leq 40$ **[B1]**

**(ii)** New $\sum fm = 1350 + 10(35) = 1700$ **[M1]**

$\dfrac{1700}{60} = \dfrac{85}{3}$ **[M1]**

$28\dfrac{1}{3}$ **[A1]**

*[Total Marks: 12]*

**7 (a)** $30 < h \leq 40$ **[B1]**

**(b)** Uses midpoints $15, 25, 35, 45, 55$ **[M1]**

$\Sigma fx = (15 \times 12) + (25 \times 25) + (35 \times 28) + (45 \times 11) + (55 \times 4)$ or
$180 + 625 + 980 + 495 + 220 = 2500$ **[M1]**

$\dfrac{2500}{80}$ **[M1]**

31.25 **[A1]**

*[Total Marks: 5]*

**8 (a) (a)** $20 < h \leq 30$ **[B1]**

**(b)** Uses midpoints: $5, 15, 25, 35, 45$ **[M1]**

$\Sigma fx = (5 \times 4) + (15 \times 12) + (25 \times 20) + (35 \times 10) + (45 \times 4)$ **[M1]**

$\Sigma fx = 20 + 180 + 500 + 350 + 180 = 1230$ **[M1]**

$\frac{1230}{50} = 24.6$ **[A1]**

*[Total Marks: 5]*

**9** **(a)** $100 < m \leq 110$ **[B1]**

**(b)** Midpoints used: $85, 95, 105, 115, 125$ **[M1]**

$\Sigma fx = 85 \times 8 + 95 \times 22 + 105 \times 31 + 115 \times 14 + 125 \times 5$ **[M1]**

$\Sigma fx = 680 + 2090 + 3255 + 1610 + 625 = 8260$ **[M1]**

$\text{Mean} = \frac{8260}{80} = 103.25$ **[A1]**

**(c)** The exact masses of individual apples are not known **[B1]**

The calculation uses the midpoints of the class intervals, assuming the data is evenly distributed within each class **[B1]**

*[Total Marks: 7]*

**10** **(a)** $30 < t \leq 40$ **[B1]**

**(b)** Midpoints: $15, 25, 35, 45, 55$ **[M1]**

$\Sigma fx = 15(14) + 25(28) + 35(32) + 45(18) + 55(8)$ **[M1]**

$= 210 + 700 + 1120 + 810 + 440 = 3280$ **[M1]**

$\frac{3280}{100} = 32.8$ **[A1]**

**(c)** $18 + 8 = 26$ **[M1]**

$26$ **[A1]**

**(d)** $40 < t \leq 50$ **[B1]**

*[Total Marks: 8]*

**11** **(a)** $140 \leq m < 160$ **[B1]**

**(b)** Mid-interval values: $110, 130, 150, 170, 190$ **[B1]**

$\sum fm = 10(110) + 22(130) + 28(150) + 14(170) + 6(190) = 11680$ **[M1]**

$\frac{11680}{80}$ **[M1]**

$146$ **[A1]**

**(c)** Total mass of second crate $= 40 \times 155 = 6200$ **[M1]**

$\frac{11680 + 6200}{120} = \frac{17880}{120}$ **[M1]**

$149$ **[A1]**

**(d)** $140 \leq m < 160$ **[B1]**

*[Total Marks: 9]*

**12** **(a)** $p + q = 25$ **[B1]**

$\frac{10p + 420 + 800 + 70q + 450}{60} = 48$ or $10p + 70q = 1210$ [B2] (Allow B1 for unsimplified $\Sigma fx = 10p + 420 + 800 + 70q + 450$ seen)

Correctly sets up and solves simultaneous equations to eliminate one variable **[M1]**

$p = 9$, $q = 16$ **[A1]**

**(b)** Identifies median is between 30th and 31st value and shows cumulative frequencies 23 and 39 (or $9 + 14 = 23$ and $23 + 16 = 39$) **[M1]**

$40 < m \leq 60$ **[A1]**

*[Total Marks: 7]*

**13** **(a)** Since the mode is 8, there must be at least two 8s. **[M1]**

Since the median is 11, the middle number when ordered is 11. The first three numbers are $8, 8, 11$. **[M1]**

The mean is 12, so the sum of the five numbers is $5 \times 12 = 60$. **[M1]**

The range is 15, so the largest number is $8 + 15 = 23$. **[M1]**

Let the fourth number be $x$. Then $8 + 8 + 11 + x + 23 = 60 \implies 50 + x = 60 \implies x = 10$. **[A1]**

**(b)** New sum $= 6 \times 13 = 78$ **[M1]**

$k = 78 - 60 = 18$ **[A1]**

*[Total Marks: 7]*

**14** **(a)** **(a)** Sets up equation for the mean: $\frac{(x-3) + x + (x+2) + 2x + (2x+5) + 3x}{6} = 9$ **[M1]**

Simplifies numerator: $10x + 4 = 54$ **[M1]**

$10x = 50 \implies x = 5$ **[A1]**

**(b)** Substitutes $x = 5$ to find the ordered list: $2, 5, 7, 10, 15, 15$ **[M1]**

Lower quartile $= 5$ and Upper quartile $= 15$, so IQR $= 15 - 5 = 10$ **[A1]**

**(c)** Recognises that for 7 numbers, the median is the 4th value in the ordered list. **[M1]**

The current values are $2, 5, 7, 10, 15, 15$. For the 4th value to be 8, $y$ must be placed exactly at the 4th position, giving $2, 5, 7, 8, 10, 15, 15$. Thus $y = 8$. **[A1]**

*[Total Marks: 7]*

**15** **(a)** $15 + x + y + 8 + 2 = 50$ leading to $x + y = 25$ **[B1]**

**(b)** Midpoints: $1, 3, 5, 7, 9$ **[M1]**

$\dfrac{1(15)+3(x)+5(y)+7(8)+9(2)}{50} = 3.8$ **[M1]**

$15+3x+5y+56+18 = 190$ **[M1]**

$3x+5y+89 = 190$ leading to $3x+5y = 101$ **[A1]**

**(c)** $3x+3y = 75$ or substitution method $3(25-y)+5y = 101$ **[M1]**

$2y = 26 \implies y = 13$ **[A1]**

$x = 12$ **[A1]**

**(d)** $2 < m \leq 4$ **[B1]**

*[Total Marks: 9]*